



Arab Academy for Science, Technology and Maritime Transport

College of Computing and Information Technology

DEPARTMENT: CYBERSECURITY

COURSE: OPERATING SYSTEMS

ASSIGNMENT: OS LAB PROJECT

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The Script:

os.sh

```
#!/bin/bash

if ! command -v pdftotext &> /dev/null; then
    echo "Error: pdftotext is not installed. Please install it using 'sudo apt install poppler-utils'."
    exit 1
fi

read -p "Enter the path to your transcript PDF: " transcript_pdf
read -p "Enter the path to your department's course plan PDF: " course_plan_pdf

if [[ ! -f "$transcript_pdf" ]]; then
    echo "Error: Transcript file not found at $transcript_pdf"
    exit 1
fi

if [[ ! -f "$course_plan_pdf" ]]; then
    echo "Error: Course plan file not found at $course_plan_pdf"
    exit 1
fi

echo "Ensure you have the correct files as you are responsible for obtaining your transcript and course plan."

output_file="Course_Analysis.txt"

pdftotext "$transcript_pdf" transcript_raw.txt

declare -A completed_courses

current_code=""
while IFS= read -r line; do

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current_code=""
while IFS= read -r line; do
    if [[ $line =~ ^[A-Z]{3}[0-9]{4}$ ]]; then
        current_code="$line"
    elif [[ $line =~ ^[A-F][+]?$|^ [A-F]$|^U$ ]]; then
        if [[ -n $current_code ]]; then
            completed_courses["$current_code"]="$line"
            current_code=""
        fi
    fi
done < transcript_raw.txt

course_plan_courses=$(pdftotext "$course_plan_pdf" - | grep -oP '[A-Z]{3}\d{4}' | sort | uniq)

courses_to_be_taken=$(comm -23 <(echo "$course_plan_courses") <(printf "%s\n" "${!completed_courses[@]}" | sort | uniq))

{
    echo "Completed Courses (with Grades):"
    for code in "${!completed_courses[@]}; do
        printf "%-10s %s\n" "${code}" "${completed_courses[$code]}"
    done
    echo ""
    echo "Courses to be Taken:"
    echo "$courses_to_be_taken"
} > "$output_file"

echo "Analysis complete. Results saved to $output_file."
```

The Output:

```
mohamed@mohamed-elsayed: ~/Desktop
mohamed@mohamed-elsayed:~/Desktop/OSlab$ ./os.sh
Enter the path to your transcript PDF: /home/mohamed/Desktop/OSlab/StudentTranscript.pdf
Enter the path to your department's course plan PDF: /home/mohamed/Desktop/OSlab/CY June 2024.pdf
Ensure you have the correct files as you are responsible for obtaining your transcript and course plan.
Analysis complete. Results saved to Course_Analysis.txt.
mohamed@mohamed-elsayed:~/Desktop/OSlab$
```

Completed Courses (with Grades):

UNR1302	A+
EBA1203	A+
EBA1204	A+
CCY3101	U
CCY3002	U
CAI3101	U
CCS2401	A+
CCS2201	A+
CCS2303	A
CCS2305	A
CCS2304	A
CCS2103	A+
CCS2102	A
CIS1000	A-
CCS1001	A-
CCS1302	A
CCS3203	U
UNR1403	C
UNR1407	B
CIT3200	U
CCY2001	B+
CIS2101	A
EBA1110	A+

EBA1106	A
CCS1101	A+
CNC1401	B+

Courses to be Taken:

CCS3403
CCY3003
CCY3102
CCY3201
CCY3302
CCY4202
CCY4301
CCY4303
CCY4401
CCY4901
CCY4902
EBA0201
GLA0001